

Efficient Energy Consumption in Wired Networks

BSC in Computing (Network)
COMP1665: Advanced Networks
University of Greenwich
Old Royal Naval College
United Kingdom

Abstract — Energy efficiency in backbone network is a concern with the exponential increase of internet traffic, and energy proportional computing (where energy consumption scales with traffic) offers optimal solution to reducing energy waste. This paper explores energy reduction techniques, analysing energy consumptions based on router states, such as idle, sleep, receive or transmit, and work on optimization strategies using NetSim simulations. Findings in this paper suggest energy proportional routing, and sleep scheduling can significantly reduce consumption without impacting the network performance.

Key words: network energy consumption, routing states, NetSim Simultaion, energy proportional computing,

I. Introduction

Drastic growth of online communication resulted in increased energy consumption in backbone network. Traditional routes have constant energy consumption regardless of the traffic load, leading to energy inefficiencies and waste. According to (Can Eyupoglu, 2015), most energy is consumed by network devices during idle state. Approaches in literature to address this issue could be divided into two categories; energy proportional routing, and sleep mode approaches.

According to (International Energy Agency (IEA), 2022), energy usage of network devices and data center account for 1-1.5% (240-340 TWh) of global energy usage. In this paper, energy consumptions based on router states (such as idle, sleeping or operating) are analyzed, and optimization strategies are explored using NetSim simulation (Wood, 2017). This paper will include a literature review on energy efficient routing, simulation study in NetSim, and analysis of energy usage in different circumstances, and traffic conditions.

II. Estimating Energy Consumption

There are several models to calculate energy consumption; such as the widely cited **energy proportional computing** model by (Hölzle, 2007), where power usage is directly proportional to utilization, or the **Packet-level energy model** (R. Ratheesh, 2023).

However, in this paper, following hybrid model will be used to for energy estimation as it balances accuracy, feasibility and aligns with energy-proportionality principles.

Total Energy Consumed for one router

$$\begin{aligned} &= P_{idle} * T_{idle} \\ &+ P_{sleep} * T_{sleep} \\ &+ \left(\sum_z (N_{receive}(z) * P_{receive}(z)) \right) * T_{receive} \\ &+ \left(\sum_z (N_{transmit}(z) * P_{transmit}(z)) \right) * T_{trasmit} \end{aligned}$$

This formula is simple yet effective way to estimate router energy consumption, with assumptions such as

- routers consume energy when inactive
- sleep mode may reduce energy consumption at the cost of increased latency
- ideal router consumes power proportional to traffic (Hölzle, 2007)

The formula ignores factors such as queuing, processing overhead and cooling.

A. Energy efficient routing protocol

Factors such as convergence time, computational complexity, or routing update frequency may indirectly impact the consumption. Energy aware routing (EAR) is an efficient strategy to optimize consumption, and fully compatible with the OSPF (A. Cianfrani, 2011). An ideal green routing protocol in this regard is the Energy Aware OSPF (EA-OSPF) as it includes energy consumption when calculating the link-cost metric, allowing under utilized devices to enter low-power state.

However, while NetSim may not support EA-OSPF, green energy routing could still be achieved by manually adjusting the metrics in standard OSPF. Since, EA-OSPF, is an extension of OSPF, it will work seamlessly in the OSPF enabled network without requiring any infrastructure changes.

B. Smart Sleeping for Energy-Efficiency

Smart Sleeping is a technique to reduce the energy consumption of network devices by dynamically reducing power draw from underutilized network components (such as routers) during low traffic without impacting the network performance. This approach is useful for Energy Aware Routing (EAR) and Green Networking (L. Chiaraviglio, 2009).

For example, when a link or device is underutilized, it is flagged for potential sleep. Protocols such as EA-OSPF is re-routed to consolidate the traffic, and the un-used devices are put to sleep. When traffic increases, sleeping devices are re-activated using fast-wakeup mechanisms to reduce delay (A. P. Bianzino, 2012).

III. Critical evaluation

Following simplified version of GÉANT network is created due to the limitation of NetSim simulator; consisting of 9 cities, the network interconnects European institutes in cities such as London, Paris, etc. with fibre links.

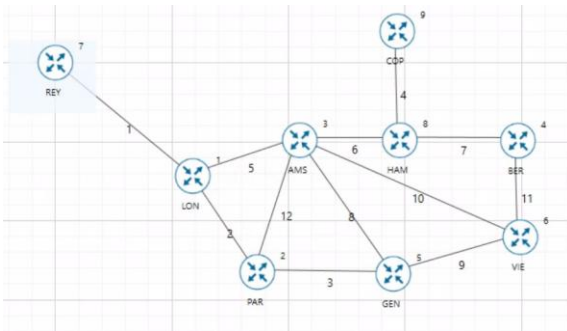


Fig 3.1 Simplified version of GEANT network in NetSim simulator

Cities in the GEANT network are represented by routers, with OSPFv2 as the routing protocol with 1800 second LS Refresh Time.

Low Traffic

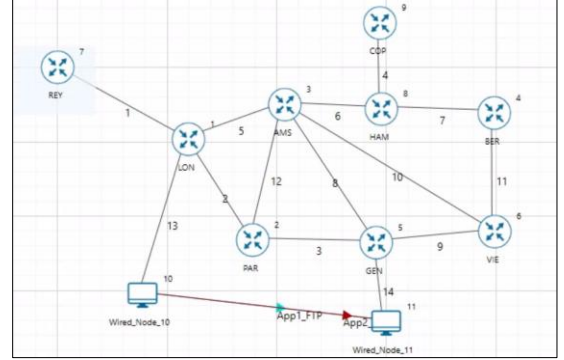


Fig 3.2 FTP and HTTP traffic between LON and GEN

For low traffic network, **HTTP and FTP traffic** with packet size of 300,000 bytes is sent between LON and GEN, travelling along the path of LON>PAR>GEN.

Application Name	Source ID	Destination ID	Throughput (Mbps)	Delay (µs)	Jitter (µs)	Packets Generated	Packets Received
App1_FTP	10	11	0.160000	506842.153043	7535.469781	138	138
App2_HTTP	11	10	0.005400	467.768889	243.200000	9	9
App2_HTTP	10	11	0.144000	57104.929524	8235.791360	126	126

Fig 3.3 Low Traffic condition Application metric between LON and GEN

From Fig 3.2, we observed Router 11 (GEN) receives around 138 packets for FTP application, with delay of approximately .5 seconds (500,000 microseconds).

If we assume energy consumption per packet = 0.1W, P idle = 0.05W/second, P sleep = 0.01W/second for 10 second idle-time and sleep like 30 seconds for this instance, we get 0.5J of *idle consumption*, 0.3J of *sleep consumption*, 6.9J of energy for receiving the consumption, and no energy consumed for transmission.

Using the hybrid-power estimation formula chosen in section II, we get

Total energy consumption = 0.5 + 0.3 + 6.9 = 7.7J for one router

For the HTTP traffic, the total packet transmitted and received on one router is 135 (close to 138) on the FTP traffic, thus we could assume the total energy consumption for this low traffic condition (FTP+HTTP) to be 7.7J + 7.7J = 15.4J for one router.

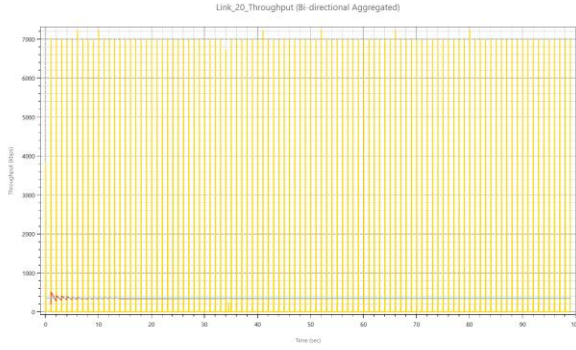


Fig 3.4 Metric Graph for Low-Traffic condition

High Traffic

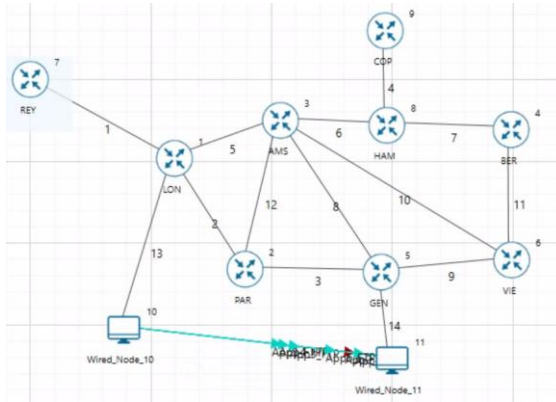


Fig 3.5 High FTP and HTTP traffic between LON and GEN

As for the high traffic condition, **nine FTP and HTTP application** traffic is placed between LON and GEN as shown in Fig 3.5.

Application Name	Source ID	Destination ID	Throughput (Mbps)	Delay (μs)	Jitter (μs)	Packets Generated	Packets Received
App1_FTP	10	11	0.160000	525486.060290	8350.493431	138	138
App2_HTTP	11	10	0.005400	835.608889	1067.960000	9	9
App2_HTTP	10	11	0.144000	5318.998095	1119.708160	126	126
App3_FTP	10	11	0.160000	1015325.357681	15050.070657	138	138
App4_FTP	10	11	0.160000	525206.188986	8189.187737	138	138
App5_FTP	10	11	0.160000	536751.820290	8338.937226	138	138
App6_FTP	10	11	0.160000	529279.016812	8287.084380	138	138
App7_FTP	10	11	0.160000	531649.952464	8342.231825	138	138
App8_FTP	10	11	0.160000	534065.108406	8404.837372	138	138

Fig 3.6 High-Traffic condition Application metric between LON and GEN

We assume the same amount of energy is consumed per packet, and the router has 10 second idle-time. As shown in Fig

3.4, we have total of seven FTP traffics (each with 138 packets and around .5 second delay).

By using the hybrid power estimation model, $7 \times 7.7J = 53.9J$, and one HTTP traffic, also around $7.7J$. Making the total power consumption of router = $61.6J$.

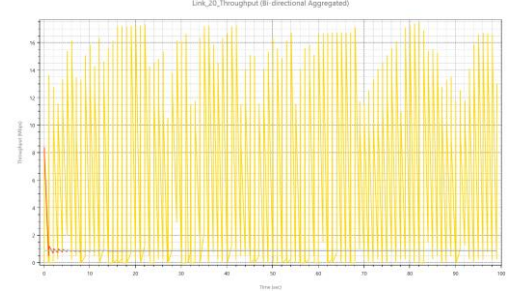


Fig 3.7 Throughput graph High-Traffic condition

Smart Sleeping

If we include the Smart Sleeping feature in the high traffic scenario, flagging the underutilized routers for sleep, it will consume less power. With the Smart Sleeping, we will replace P idle in the hybrid formula with P sleep. As assumed above, we have

$P_{sleep} = 0.01W$ and $P_{idle} = 0.05W$ during the 10 second idle period (smart sleeping), $P_{sleep} = 0.01W \times 10 \text{ seconds} = 0.1J$, while for $P_{idle} = 0.05W \times 10 \text{ seconds} = 0.5J$.

For eight HTTP and FTP applications, we saved $(0.5 - 0.1) \times 8 = 3.2J$, saving around 5.2% energy. Thus, final total energy consumption with Smart Sleeping is

$$61.6J - 3.2J = 58.4J$$

IV. Conclusions

This paper explores energy consumption in wired backbone network, by analyzing the simplified model of GEANT network in NetSim simulator. The analysis highlighted the importance of power-proportional computing, and Energy Aware technology (such as EA-OSPF routing protocol) to reduce power usage of network devices, without impacting the network performance. It is also discovered that simply implementing Smart Sleeping feature could reduce around 5.2% of energy consumption, saving around 12.5-17.7 TWh per year globally.

Acknowledgement

I wish to thank Dr. Aung Htein Maw for guiding and providing us necessary knowledge in the Networking field. In

addition, I also wish to thank my parents, family and friends for encouraging and helping me throughout this journey.

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